

# Songning Zhu

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Ph.D. in Biomedical Engineering with over five years of research experience designing and integrating advanced optical instrumentation for high-precision photonic measurements and imaging. Developer of PRISMOS, a phasor-based hyperspectral imaging methodology (patent pending). Skilled in multi-wavelength laser systems, interferometric signal detection, spectral unmixing, and computational signal processing using deep learning. Proven track record of interdisciplinary collaboration, high-impact publications, and extracting quantitative information from high-dimensional optical datasets.

## EDUCATION

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| <b>Stanford University, Palo Alto, Department of Applied Physics</b><br><i>Postdoctoral Research Scholar</i>                                                          | April 2026 - Present                           |
| <b>University of California, Irvine, The Henry Samueli School of Engineering</b><br><i>Doctor of Philosophy in Biomedical Engineering, Biophotonics Concentration</i> | December 2025<br>GPA:3.96/4.00                 |
| <b>The Pennsylvania State University, University Park - College of Engineering</b><br><i>Bachelor of Science in Biomedical Engineering</i><br><i>Cum Laude</i>        | State College, PA<br>May 2020<br>GPA:3.83/4.00 |

## TECHNICAL SKILLS:

**Scientific Computing:** Python, MATLAB, R, ImageJ, SOLIDWORKS, Microsoft Office

**Optical & Photonic Systems:** Multi-wavelength laser systems, fluorescence lifetime and spectral detection, hyperspectral acquisition, photon-counting detectors, beam alignment and stabilization, interferometric signal processing, fiber coupling and alignment, optical system calibration (ZEISS LSM series, Evident, custom-built platforms)

**Data Processing & Machine Learning:** PyTorch, Pandas, NumPy, scikit-learn (sklearn)

**Laboratory Techniques:** DNA Purification/Amplification, ECG Amplifier Design, Spin-Coating, Graphene Monolayer Exfoliation, FESEM, Block Copolymer Synthesis

**Cell Culture:** Human Cell Line, Animal Cell Line, MTT assay, transfection

## EQUIPMENT MAINTENANCE

**Microscopy Systems:** Regular maintenance and calibration of advanced microscopy equipment: ZEISS LSM 880, LSM 710, ISS Alba, and home-built bioluminescence imaging system

**Lab Equipment:** Oversaw the operation and upkeep of 4 incubators and 5 biosafety cabinets, ensuring compliance with laboratory safety protocols.

## RESEARCH EXPERIENCE:

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|---------------------------------------------------------|--------------------------------------------|
| <b>Digman Lab</b><br><i>Graduate Research Assistant</i> | Irvine, CA<br>July 2020 till December 2025 |
|---------------------------------------------------------|--------------------------------------------|

- Developed **PRISMOS** (Phasor-Resolved Hyperspectral Imaging with Single Excitation), an advanced imaging platform for **multiplexed organelle phenotyping** in chemoresistant triple-negative breast cancer cells.
- Integrated **confocal, two-photon, and FLIM-based hyperspectral imaging** with **phasor unmixing models** to visualize subcellular dynamics at high spatial resolution.
- Collaborated with physics and AI researchers to implement **stochastic, physics-informed neural networks** for live-cell imaging data compression and prediction.
- Developed and evaluated mechano-sensitive biomaterials for cardiomyocyte imaging using hyperspectral and time-domain signal analysis.

## Bio-Photonics and Ultrasound Imaging Laboratory

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|-----------------------------------------|----------------------------------------------------|
| <i>Undergraduate Research Assistant</i> | State College, PA<br>January 2019 to December 2019 |
|-----------------------------------------|----------------------------------------------------|

- Understanding and operation of the functionality of photo-acoustic imaging: ACOUSTIC X, Verasonics
- Investigation of photo-acoustic and ultrasounds imaging in the application of neonatal brain
- Modeling the biological response to photo-acoustic stimulus using MATLAB and SOLIDWORKS

## PUBLICATIONS

Scipioni, L., Tedeschi, G., Navarro, M., Jia, Y., **Zhu, S.**, Halbers, L. P., Di Bona, M., Atwood, S., Prescher, J. A., Gratton, E., & Digman, M. *ESPRESSO: Spatiotemporal omics based on organelle phenotyping*. *Nature Methods* (Accepted, In Press).

Hedde, P. N., **Zhu, S.**, Barylko, B., Chiu, C. L., Nelson, L. T., Digman, M. A., Albanesi, J. P., James, N. G., & Jameson, D. M. (2024). Effect of pathogenic mutations on the formation of high-order Dynamin 2 assemblies in living cells. *Biochemistry*, 63(21), 2750-2758. [DOI: 10.1021/acs.biochem.4c00262]. PMID: 39390788.

Fazel, M., Jazani, S., Scipioni, L., Vallmitjana, A., **Zhu, S.**, Gratton, E., Digman, M. A., & Pressé, S. (2023). Building fluorescence lifetime maps photon-by-photon by leveraging spatial correlations. *ACS Photonics*.

**Zhu, S.**, Digman, M., Thompson, J., & Campos-Chillon, F. (2023). IVF bovine oocyte classification and selection. *Biophysical Journal*, 122(415a). [DOI: 10.1016/j.bpj.2022.11.2253].

**Zhu, S.**, Scipioni, L., & Digman, M. (2022). Phasor unmixing to reveal organelle organization and cellular response. *Biophysical Journal*, 121(278a). [DOI: 10.1016/j.bpj.2021.11.1352].

Manuscripts in Preparation: **Zhu, S.**, Scipioni, L., & Digman, M. PRISMOS: Single-Excitation Phasor-Resolved Hyperspectral Imaging for Multiplexed Organelle Phenotyping in Chemoresistant Triple-Negative Breast Cancer (main PhD. thesis)

## PROFESSIONAL EXPERIENCE:

**Optical Engineer Intern, KLA** Milpitas

- Investigate different design options and finding the best option through evaluating tolerances, manufacturability, light budgets and other trade-offs.
- Perform a tolerancing analysis for optical components or optical system
- Building analysis pipeline with computer science department for precision control

Consultation for System building and design

**Capstone Design Project, BME Department at Pennsylvania State University** State College, PA  
*Project Leader* Fall 2019

- Facilitated meetings, delegated tasks, and ensure completion of the team's mission to design a micro-fluidic device that allows plasma protein isolation with a budget of \$3000 over the course of a semester
- Manufactured the graphene oxide monolayer member via block co-polymer synthesis and reactive ion etching. And the characterization via FESEM
- Compiled a 41 page proposal and a showcase poster to the sponsor company.
- Effectively communicated with the other team for a smooth transition of the remaining project

## CAMPUS & COMMUNITY INVOLVEMENT

**Undergraduate Student native for Biomedical Research (USIBR)** July 23 - July 28, 2023 & July 29 - August 2024  
*Workshop leader*

- Lecturing & Teaching: Deliver engaging lectures on topics related to optical imaging and quantitative biology alongside Prof. Digman, Prof. Enrico Gratton, and other expert postdoctoral fellows.
- Mentorship & Leadership: Lead a team of 5-10 undergraduate students through an intensive, hands-on research experience, fostering critical thinking and hypothesis-driven exploration.
- Project Guidance: Design and guide "mini" research projects, allowing students to test real-world biological questions and contribute to ongoing research in the lab.
- Skill Development: Help students develop laboratory techniques, data analysis skills, and scientific communication abilities in a collaborative and supportive environment.

**Vice President of Social Affairs, AGS** August 2021 till August 2022

- Event planning & Execution: Lead the social committee in organizing monthly pub bights, trivia/karaoke events and large-scale social gatherings
- Campus-wide coordination: represented graduate student interests in major campus events, including graduate student orientation
- Marketing & Outreach: Collaborated across different organizations such as SWE, Partnered with organizations such as the Graduate Resource Center (GRC), International Center, DECADE, GPS-STEM, and departmental graduate associations to co-host events and expand outreach.
- Leadership: Worked with AGS executives and council members to align social programming with student need

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| <b>Social Program Coordinator, AGS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | August 2021 till August 2022  |
| <ul style="list-style-type: none"> <li>Design, organize and conduct UCI graduate student community-based social programs and service</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                |                               |
| <b>International Chair, AGS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | August 2020 to August 2021    |
| <ul style="list-style-type: none"> <li>Identify and assess the needs and desires of international graduate students</li> <li>Design and provide quarterly events of a social, professional development, or networking nature and coordinate with the clubs and resources on campus.</li> <li>Communicate with service providers present on campus about international students.</li> <li>Advocate for international student needs and concerns both on and off-campus.</li> <li>Create a gateway for international student participation in the Student Government.</li> </ul> |                               |
| <b>Teaching Assistant for Biomechanics</b> , University of California, Irvine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | December 2020 to March 2021   |
| <ul style="list-style-type: none"> <li>Held study sessions to go over important concepts in class</li> <li>Meeting with the instructor for student feedbacks</li> <li>Grade the weekly homework and exams</li> </ul>                                                                                                                                                                                                                                                                                                                                                           |                               |
| <b>Organic Chemistry Tutor</b> , The Pennsylvania State University                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | January 2018 to December 2018 |
| <ul style="list-style-type: none"> <li>Held study sessions and discussion session for study assist and exam preparation</li> <li>Collaborated and assisted the teaching team to arrange academic plans and provide student feedbacks.</li> </ul>                                                                                                                                                                                                                                                                                                                               |                               |
| Undergraduate Research Symposium (UROP) Oral Presentation Judge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | May 2024                      |
| <ul style="list-style-type: none"> <li>UROP presentation video competition Judge</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                               |
| Undergraduate Research Symposium (UROP) Research Symposium Judge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | May 2024                      |
| Undergraduate Research Symposium (UROP) Poster Presentation Judge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | May 2024                      |
| Undergraduate Research Symposium (UROP) Research Symposium Judge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | December 2024                 |

## AWARDS

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|-----------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Henry Samueli Endowed Fellowship, UC Irvine</b>                                                                                            | 2025 |
| Awarded for academic excellence and research contributions in biomedical engineering                                                          |      |
| <b>Recipient, Center for Complex Biological System</b>                                                                                        | 2024 |
| "Learning from live cell hyper spectral fluorescence imaging data via stochastic physics informed equivariant autoencoders" (TOTAL: \$11,000) |      |
| <b>Undergraduate Research Symposium (UROP) Awardee</b>                                                                                        | 2023 |
| "Evaluation of anti-breast cancer treatments at the organelle level" (TOTAL: \$400)                                                           |      |
| <b>Second Place, AGS Annual Graduate Research Symposium</b>                                                                                   | 2023 |

## MEETING & CONFERENCES

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| <b>Biophysical Society 66<sup>th</sup> Annual Meeting</b>                                                                                | <b>San Francisco</b> |
| Student presenter                                                                                                                        | February 2022        |
| Presentation topic: Phasor Unmixing to Reveal Organelle Organization and Cellular Response. Zhu, Songning et al.                         |                      |
| Biophysical Journal, Volume 121, Issue 3, 278a.                                                                                          |                      |
| <b>Center for Advanced Design and Manufacturing of integrated Microfluidics Meeting</b>                                                  | <b>Irvine</b>        |
| Presentation topic: Phasor Unmixing to Reveal Organelle Organization and Cellular Response.                                              | February 2022        |
| <b>Biophysical Society 67<sup>th</sup> Annual Meeting</b>                                                                                | <b>San Francisco</b> |
| Student presenter                                                                                                                        | February 2023        |
| Presentation topic: IVF Bovine Oocytes Classification and Selection. Zhu, Songning et al. Biophysical Journal, Volume 122, Issue 3, 415a |                      |
| <b>Biophysical Society 68<sup>th</sup> Annual Meeting</b>                                                                                | <b>San Diego</b>     |
| Student presenter                                                                                                                        | February 2024        |
| Presentation topic: IVF Bovine Oocytes Classification and Selection. Zhu, Songning et al. Biophysical Journal, Volume 122, Issue 3, 415a |                      |

**Biophysical Society 69<sup>th</sup> Annual Meeting**  
Student presenter

**Los Angeles**  
February 2025

**15th - 18th LFD Workshop in Advanced Fluorescence Imaging and Dynamics**

Workshop Leader: Leading hands-on session for over 30 participants  
Workshop Lecturer: Lecture Hyperspectral Imaging

**Irvine**  
October 2021-2024

**MEDICAL SHADOWING**

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**UCI hospital**

Radiology/Advanced Imaging, Pathology, Advanced Heart Failure, Nephrology, Gastroenterology, Vascular Surgery, Urology  
5-7 hours/department ~40 hours

**Xiangya Hospital Central South University Psychiatry department**

**July 2019**

- Laboratory of National Clinical Research Center for Mental Disorders 45 hours
- Under the supervision of Dr. Hao Wei.

**China's National Center for Cardiovascular Diseases**

- Pediatric department 30 hours
- Under the supervision of Dr. Li Hanmei

**VOLUNTEERING**

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**Kaiser Permanente**

**Irvine**

**Health scholar - 300 hours**

March 2025 till December 2025

- **Clinical Support:** Work alongside nurses, physicians, and allied health professionals to assist in providing basic patient care.
- **Administrative Exposure:** Learn about healthcare operations, hospital workflows, and the business of healthcare.